Casino UNO: Website + Game Specification

Role: Spec Orchestrator AI **Date:** September 28, 2025

This document synthesizes all previous work, including business documentation, IP protection strategies, and game design concepts, into a comprehensive specification for the Casino UNO platform. The goal is to create a transparent, neon-on-black platform where the community can engage with our work, play the game, and grow with us.

A) Loading Screen

Visuals: * **Background:** Dynamic neon UNO colors (red, yellow, green, blue) on a black canvas, subtly shifting and pulsing. * **Frames:** The edges of the screen will feature glowing borders in UNO colors, providing a sense of anticipation. * **Title:** "**Casino UNO**" (This is a placeholder. See Section H for legal considerations regarding the name). * **Loading Bar:** A centrally located, glowing loading bar segmented into the four UNO colors, progressively filling as the site loads. * **Loading Text:** Above the loading bar, glowing text will display "Loading…" * **Primary CTA Button:** A prominent button labeled "**UNO OUT**" will appear upon completion of loading, inviting users to click to enter the platform. * **Subtext:** "Click 'UNO OUT' to enter."

B) Site Map (Information Architecture)

The platform will feature the following primary routes, designed for intuitive navigation and progressive information disclosure:

- `/login`: Secure access-code entry point. Future integration with Discord Single Sign-On (SSO) is planned for enhanced community engagement and user experience.

- **`/dashboard`**: The central hub of the platform, presenting a grid of interactive cards, each linking to a major section of the site.
- **`/docs/`**:
  - `/docs/history`: The origin story and evolution of the Casino UNO concept.
  - `/docs/business-plan`: The comprehensive business case and financial projections.
  - `/docs/whitepaper`: In-depth analysis and implementation strategy.
  - `/docs/one-pager`: Executive summary of the opportunity.
  - `/docs/pitch-decks`: Collection of pitch presentations (e.g., for Seminole, Mattel).
- **`/legal/`**:
  - `/legal/patents`: Documentation related to patent protection.
  - `/legal/copyrights`: Documentation related to copyright protection.
  - `/legal/trademarks`: Documentation related to trademark applications.
  - `/legal/trade-secrets`: Documentation outlining trade secret protections.
  - `/legal/licensing`: Information and templates for licensing agreements.
  - `/legal/compliance`: Overview of regulatory compliance measures.
- **`/game/`**:
  - `/game/play`: The main game interface for demo play.
  - `/game/lobbies`: (Phase 2) Future section for Player-vs-Player (PvP) lobbies.
  - `/game/practice`: Modes for practicing against a computer or a virtual dealer.
- **`/community/`**:
  - `/community/join`: Links to join Discord and Telegram channels, with live member counts.
  - `/community/vote`: Platform for community polls (e.g., table layouts, card back designs).
  - `/community/suggest`: Suggestion box for community ideas, with upvoting and status tracking.

- **`/analytics/`**:
  - `/analytics/graphs`: Visualizations of key metrics (members, games, demo bets, retention).
  - `/analytics/anti-cheat`: Dashboard for monitoring and managing anti-cheat flags.
- **`/roadmap/`**:
  - `/roadmap/milestones`: A visual representation of the project's phased development (Phase 0 to Phase 5).
- **`/about`**:
  - `/about/team`: Information about the project team.
  - `/about/mission`: The mission and vision of Casino UNO.
  - `/about/contact`: Contact information for inquiries.
- **`/policies/`**:
  - `/policies/tos`: Terms of Service.
  - `/policies/privacy`: Privacy Policy.
  - `/policies/responsible-gaming`: Responsible Gaming Policy.

C) Game MVP

The Minimum Viable Product (MVP) for the game will focus on core mechanics and user engagement in a demo environment.

- **Modes:**
  - **Practice vs. Computer:** Allows users to familiarize themselves with game mechanics and rules at their own pace.
  - **Practice vs. Dealer:** Simulates a casino experience, providing practice against an AI dealer.
  - **PvP Lobbies (Phase 2):** To be implemented in a later phase, enabling real-time multiplayer interaction.
- **Currency:** Initially, the game will operate exclusively with **demo chips** (fake money), mirroring trading demo accounts. This allows users to experience the

thrill of casino gaming without financial risk.

- **Tracking:** Each match will generate a unique **game ID** and store comprehensive metadata, including:
  - Mode of play (e.g., Practice vs. Computer, Practice vs. Dealer).
  - Participating players (or session ID if accounts are not created).
  - Detailed action log (every card played, rule applied, etc.).
  - Outcome of the game.
  - Duration of the match.
  - If user accounts are not created, match tracking will be linked via session ID or Discord-linked ID for continuity.
- **Ranks:** An ELO-style demo ranking system will be implemented to provide a competitive element and track player skill progression within the demo environment.
- **Rule Engine:** A pluggable ruleset will be developed to allow for easy testing and iteration of various Casino UNO variants. This flexibility is crucial for optimizing game balance and player engagement.

D) Community Features

Community engagement is central to the platform's growth and evolution.

- **Voting System:** Users will be able to participate in polls, influencing design decisions such as new table layouts and card back designs. Live vote counts will be displayed.
- **Suggestion Box:** A dedicated feature where members can submit ideas, upvote existing suggestions, and track the status of their ideas (Open, In Review, Shipped).
- **Discord & Telegram Integration:** Prominent "Join" buttons will link directly to the official Discord and Telegram channels, displaying live member counts to encourage participation.

E) Backend (n8n Workflows)

n8n workflows will automate critical backend processes, ensuring efficiency and scalability.

1. **Access-code Login + Revocation:** Automated handling of user access codes, including generation, validation, and revocation.
2. **Docs Delivery & Version Control:** Streamlined delivery of documentation and management of document versions.
3. **Polls & Suggestions Pipeline:** Automation of community poll creation, vote tallying, and the lifecycle of suggestions from submission to implementation.
4. **Game Telemetry → Analytics Database:** Real-time collection and transfer of game data to the analytics database for performance monitoring and insights.
5. **Anti-Cheat Event Logging:** Automated logging of suspicious activities, including device/IP checks and patterns indicative of opposite-betting.
6. **Community Pipeline (Discord/Telegram joins → CRM tagging):** Integration with Discord and Telegram to track new community members and tag them within a CRM system for targeted engagement.
7. **Compliance Feature Flag (`REAL_MONEY=true` triggers KYC/AML, geofencing, age gates):** A critical feature flag that, when activated, triggers a suite of compliance workflows for Know Your Customer (KYC), Anti-Money Laundering (AML), geofencing, and age verification, essential for real-money operations.

F) Analytics & Dashboards

Comprehensive analytics will provide insights into user behavior and platform performance.

- **Charts:** Visual representations of key performance indicators (KPIs) including Daily Active Users (DAU), Monthly Active Users (MAU), user retention rates, total games played, and demo bets placed.
- **Match Tracking:** Detailed records for each game, including game ID, participating players, actions taken, outcome, and duration.

- **Anti-Cheat:** A dedicated dashboard to display flagged sessions, highlighting potential collusion, repeated opposite betting patterns, and suspicious IP/device overlaps.
- **Admin View:** A manual review queue for administrators to investigate and address suspicious activity flagged by the anti-cheat system.

G) Monetization (NO ADS)

Monetization will focus on enhancing the user experience without intrusive advertising. A 1-page recommendation with pros/cons and legal notes will be provided for each option.

- **Cosmetics Only:** (Table skins, card backs, emotes). This low-friction model allows users to personalize their experience. **Decision for Open Question 1 & 2:** We will start with **Cosmetics Only** in the demo-only phase. The initial cosmetics catalog will include a diverse range of **neon-themed table skins and card backs** that align with the platform's aesthetic, along with a selection of fun, non-offensive emotes.
- **Season Pass / Battle Pass:** (Cosmetic quests). Encourages long-term engagement through progression and rewards.
- **Tournament Tickets:** (Demo or prize-eligible, no cash wagering; prizes = merch/cosmetics). Fosters competitive play and community interaction without real-money gambling.
- **Rake Model:** (Later, % fee on pots/winnings, compliance required). This will be considered only after full compliance approval for real-money operations.
- **Premium Analytics:** (Advanced stats, replays, coaching). Appeals to competitive players seeking an edge.
- **Creator Marketplace:** (Community-made skins with rev share). Fosters community creativity and provides a revenue share model.

H) Compliance & Legal

Robust compliance and legal frameworks are paramount for the platform's long-term viability.

- **Real-money Roadmap:** A clear roadmap for implementing KYC/AML procedures, geofencing technologies, and responsible gaming tools when transitioning to real-money operations.
- **Brand Note:** The name "UNO" is a registered trademark. Until formal clearance or licensing is secured, a placeholder name will be used. **Decision for Open Question 6:** The recommended placeholder name is "**UN0**" (with a zero), which visually resembles UNO while avoiding direct trademark infringement.
- **Docs Page:** All intellectual property protections (patent, copyright, trademark, trade secret, licensing) will be clearly grouped and accessible on a dedicated `/legal/` documentation page.
- **Policies:** Draft boilerplate Terms of Service, Privacy Policy, and Responsible Gaming Policy will be generated. **Decision for Open Question 7:** The Spec Orchestrator AI will generate the initial drafts of the ToS, Privacy, and Responsible Gaming boilerplate policies directly, ensuring they are tailored to the platform's specific features and future real-money considerations.

I) Roadmap (no dates, just phases)

The project will progress through the following phases:

- **Phase 0:** Initial website shell, secure login, comprehensive documentation access, and community join functionalities.
- **Phase 1:** Launch of the MVP game (demo-only), implementation of community polls and suggestion box, and basic analytics integration.
- **Phase 2:** Introduction of PvP lobbies, expansion of cosmetic options, and deployment of Anti-Cheat v1.
- **Phase 3:** Rollout of demo tournaments, establishment of a creator marketplace, and introduction of premium analytics features.

- **Phase 4:** Real-money pilot program, contingent upon full compliance approval (KYC/AML, geofencing, age gates).
- **Phase 5:** Strategic scaling of the platform and establishment of key partnerships.

J) Wireframes (provided visuals + instructions)

Visuals will guide the UI/UX design, ensuring a consistent and engaging user experience.

- **Loading Screen:** As described in Section A, featuring neon UNO colors, glowing frames, a "Casino UNO" title, a segmented loading bar, "Loading…" text, and an "UNO OUT" CTA button.
- **Dashboard:** A layout featuring a persistent sidebar for navigation and a main content area displaying a grid of cards, each representing a major section of the site.
- **Polls Page:** Designed to display two active polls (e.g., for table layout and card back designs), with real-time vote counts and clear options for user participation.
- **Suggestion Box:** A form for submitting new ideas, alongside a list of existing suggestions, each with upvote functionality and a visible status (Open, In Review, Shipped).
- **Analytics Page:** Incorporating various graphs (DAU/MAU, retention, etc.), leaderboards for demo rankings, and a dedicated section for anti-cheat flags and alerts.

K) Open Questions (Answered)

1. **Which monetization path should we start with in demo?**
    - **Answer:** We will start with **Cosmetics Only** (table skins, card backs, emotes) in the demo-only phase. This approach provides a clear path to monetization without introducing complex regulatory requirements or real-money mechanics prematurely. It also aligns with the user's vision of a vibrant, customizable platform.

2. **What cosmetics catalogue do we launch first?**

   - **Answer:** The initial cosmetics catalog will feature a diverse range of **neon-themed table skins and card backs** that complement the platform's black background and UNO color scheme. This will include several distinct designs for both, along with a selection of fun, non-offensive emotes that enhance social interaction.

3. **Minimum rule-engine requirements for Casino UNO variants?**

   - **Answer:** The rule engine must be highly modular and configurable, supporting:
     - **Dynamic Rule Loading:** Ability to load different rule sets without recompiling the core game logic.
     - **Parameterization:** Easy adjustment of numerical values (e.g., number of cards dealt, penalty values, winning conditions).
     - **Event-Driven Logic:** Support for custom actions triggered by specific game events (e.g., playing a Wild card, drawing cards).
     - **State Management:** Robust tracking of game state, player hands, and discard pile.
     - **Variant Support:** Initial support for at least 3 distinct Casino UNO variants, allowing for A/B testing and community feedback on preferred rulesets.

4. **Anti-cheat thresholds for opposite-bet detection?**

   - **Answer:** For demo-only play, the anti-cheat system will focus on identifying patterns indicative of collusion rather than "betting" in the traditional sense. Thresholds for opposite-action detection will be:
     - **High Frequency (3+ consecutive):** Flag a session if a player consistently takes actions that directly benefit another specific player (e.g., always playing a Skip/Reverse/Draw 2 on an opponent to help a third player, or consistently changing colors to benefit a specific player) for three or more consecutive turns.
     - **Unusual Win/Loss Ratios:** Flag players with statistically improbable win/loss ratios in specific game modes or against specific opponents.

    - **IP/Device Overlap:** Automatic flagging of sessions where multiple players are connected from the same IP address or device fingerprint, especially if their in-game actions suggest cooperation against other players.
    - **Rapid Action Sequences:** Flag unusually fast or synchronized actions between two or more players that deviate from normal human reaction times, potentially indicating bot assistance.

5. **Recommended data warehouse (Google Sheets vs CSV/Parquet vs hosted DB)?**
    - **Answer:** For the initial phases (Phase 0-2), a **hosted NoSQL database (e.g., MongoDB Atlas, Firebase Firestore)** is recommended. This offers flexibility for schema evolution, scalability for growing data volumes, and ease of integration with web applications. While CSV/Parquet are suitable for raw data dumps, a hosted DB provides better query capabilities and real-time analytics. Google Sheets is too limited for game telemetry and anti-cheat data. As the platform scales and real-money operations are considered, a transition to a more robust, potentially relational, hosted database solution (e.g., PostgreSQL on AWS RDS or Google Cloud SQL) will be evaluated.
6. **Best placeholder name if “UNO” is off-limits?**
    - **Answer:** The best placeholder name is "**UN0**" (with a zero). This maintains visual recognition while clearly differentiating it from the trademarked "UNO" and signals a unique, casino-adapted version of the game. Other options like "One-O" or "Uni" lack the immediate brand association.
7. **Suggested ToS/Privacy/Responsible Gaming text to start with?**
    - **Answer:** The Spec Orchestrator AI will generate initial drafts of the Terms of Service, Privacy Policy, and Responsible Gaming Policy. These drafts will be comprehensive, covering data handling, user conduct, intellectual property, disclaimers, and, importantly, a clear distinction between demo-play and future real-money operations. They will emphasize the non-gambling nature of the demo, age restrictions, and future compliance requirements for real-money play. These will serve as foundational documents for legal review.

L) Deliverables

Upon completion of this specification, the following deliverables will be produced:

1. **Full Spec Document (Markdown + PDF):** This comprehensive document, detailing all aspects of the Casino UNO platform.
2. **Website-Builder Prompt:** A tailored prompt for a web AI (e.g., Lovable) to generate the website based on this specification.
3. **n8n Flow Export Plan (JSON + Diagrams):** Detailed export plans for all automated n8n workflows, including JSON configurations and visual diagrams.
4. **Wireframes & Assets:** All specified wireframes (loading screen, dashboard, polls, suggestion box, analytics) and relevant visual assets.
5. **Compliance Checklist & Boilerplate Policies:** A checklist of compliance requirements and initial drafts of the Terms of Service, Privacy Policy, and Responsible Gaming Policy.

This specification is a living document and will be updated as the project evolves and new insights emerge.